Problem A. Heshen’s Account Book

Description

Jin Yong was the most famous and popular Chinese wuxia (The one who fight bad people by his Wukong i.e. Wushu and Kongfu) novelist who lived in Hong Kong. Between 1955 and 1972, he wrote 14 novels which earned him a reputation as one of the greatest and most popular Chinese writers. Over 100 million copies of his works have been sold worldwide，not including a countless number of pirated copies. Jin Yong’s works seem to have magic. Once you begin to read a novel of his, you just can’t stop until you finish it.

Last month, Jin Yong passed away at the age of 94. Many Jin Yong’s fans in PKU held a meeting to memorize him. Jin Yong’s fans always like to discuss or argue or even quarrel about whose Wukong are better among the wuxia characters of his novel. During the meeting, this happened again:

Every fans said some words like "Qiao Feng's Wukong is better than Guo Jing's". Obviously, those words may contradict each other and then cause quarrels. As a boring and girlfriendless male programmer of EECS school, you always want to make some things. So you are eager to point out the contradictions as soon as possible. That means, you want to find out the first one whose words contradict the words said by others before him.

Please note that if A is better than B, and B is better than C, then of course A must be better than C.

Input

There are no more than 15 test cases.

For each test case:

The first line is an integer n(1 <= n <=20), meaning that there are n sentences.

The following n lines are those n sentences which is in the format below:

s1 s2

This means someone said that s1's Wukong was better than s2's. Both s1 and s2 are names of Jin Yong's characters which consists of only English letters. It's guaranteed that s1 and s2 are different, and their length is no more than 30. Names are case sensitive.

Output

For each test case, print the first sentence which cause a contradiction. If there are no contradiction, print 0 instead.

Sample Input

```
2
BrokenReputation ExtinctNun
HelloLaught EnvelopeNotFlat
6
LandOverWind LonelyLight
FireMonk CutTheForest
CutTheForest LookCrazy
```

MakeFoxRush LetMeGo
HeroAunt UniqueLand
LookCrazy FireMonk

Sample Output

0
LookCrazy FireMonk

Hint

DON'T try to figure out who are those names in the sample and waste your time.

.

Problem B. Heshen's Account Book

Description

Heshen was an official of the Qing dynasty. He made a fortune which could be comparable to a whole country's wealth by corruption. So he was known as the most corrupt official in Chinese history. But Emperor Qianlong liked, or even loved him so much that was not punished during Qianlong's regime even everybody knew he was totally corrupted.

After Qianlong quit his job, his son Jiaqing took the throne. The new Emperor hated Heshen very much, so he threw Heshen into jail and awarded him death immediately after Qianlong died. Of course Jiaqing would not forget to raid Heshen's house for money --- that was the story of "Heshen fell, Jiaqing full."

Jiaqing's man got a notebook from Heshen's home which was obviously an account book.But the text of the book was written in English! Jiaqing thought all numbers in that account book should stand for money. Please find out all numbers for Jiaqing.

The text of the account book consists only lower case letters, spaces, and digits ('0' – '9'). One or several consecutive digits form a number. But Jiaqing only cared about the ACTUAL numbers among all numbers. Only if a number DOESN'T satisfy any of the conditions below, it is an ACTUAL number:

1) The character to the left of the number is a lower case letter, for example: a123
2) The character to the right of the number is a lower case letter, for example: 123b
3) The number has one or more extra leading zeros, such as 01 , 0012….

Please note that if the last character of a line is a digit, and the first character of the next line is also a digit, those two digits are considered consecutive.

Input

There are no more than 200 lines. The length of each line is no more than 1000 characters. And it is guaranteed that every number's length is no more than 18.

There may be spaces at the end of a line, and those spaces matter.

No blank lines in the input. A line consisting of only spaces is not a blank line.

Output

Print all ACTUAL numbers in a single line in the original order.

Then, count the number of ACTUAL numbers of each line, and print them. A number X only belongs to the line which contains the first digit of X.

Sample Input

```
a19 01 17b
12 bdc 13
23 14 344 bc
```

Sample Output

12 1323 14 344
0
2
2

Sample Explanation

We assume there is no spaces at the end of each line in the Sample Input.

In the sample input, the '3' at the end of the second line and the '2' at the beginning of the third line are considered consecutive, so "1323" is a number. But this number only belongs to the second line, so the third line has only 2 numbers —— 14 and 344..

Problem C. Pythagorean triple

Description

Today is Carrie's first day in Blaniel high school.

In her first math class, the teacher Mr. Jeffery tells her a well-known story of little Gauss.

When little Gauss was still at primary school, Gauss' teacher, probably to get a bit of rest for himself, asked his class to add up all the numbers from 1 to 100.

However, much to the teacher's astonishment, Gauss quickly noted the trick of adding the numbers in pairs and wrote down the answer $101 \times 50 = 5050$.

Probably because Mr. Jeffery is tired after telling the story, he asks the class to count the number of Pythagorean triples under a given integer N.

A triple (a,b,c) is called a Pythagorean triple if a,b,c are positive integers and $a^2+b^2=c^2$

Carrie should count the number of different Pythagorean triples in which c <= N.

Notice that due to the symmetric position of a and b, Pythagorean triples (a,b,c) and (b,a,c) should be counted only once.

Carrie is not as talented at math as Gauss, but she knows her boyfriend might be.

Now as Carrie's boyfriend, can you help her solve Mr. Jeffery's problem?

Input

The first line of the input is a single integer T (T <= 10) indicating the number of test cases.

Each of the following T lines contains an integer N ($1 <= N <= 10^9$), meaning that you should count the number of triples (a,b,c) in which c <= N.

Output

For each test case, output the answer in a single line.

Sample Input

2
5
20

Sample Output

1
6

Sample Explaination

All Pythagorean triples under 20 are:
(3,4,5) (5,12,13) (6,8,10) (8,15,17) (9,12,15) (12,16,20)

Problem D. Frog and Portal

Description

A small frog wants to get to the other side of a river. The frog is initially located at one bank of the river (position 0) and wants to get to the other bank (position 200). Luckily, there are 199 leaves (from position 1 to position 199) on the river, and the frog can jump between the leaves. When at position p, the frog can jump to position $p+1$ or position $p+2$.

How many different ways can the small frog get to the bank at position 200? This is a classical problem. The solution is the 201st number of Fibonacci sequence. The Fibonacci sequence is constructed as follows: $F_1 = F_2 = 1; F_n = F_{n-1} + F_{n-2}$.

Now you can build some portals on the leaves. For each leaf, you can choose whether to build a portal on it. And you should set a destination for each portal. When the frog gets to a leaf with a portal, it will be teleported to the corresponding destination immediately. If there is a portal at the destination, the frog will be teleported again immediately. If some portal destinations form a cycle, the frog will be permanently trapped inside. Note that You cannot build two portals on the same leaf.

Can you build the portals such that the number of different ways that the small frog gets to position 200 from position 0 is M?

Input

There are no more than 100 test cases.

Each test case consists of an integer M, indicating the number of ways that the small frog gets to position 200 from position 0. ($0 \leq M < 2^{32}$)

Output

For each test case:

The first line contains a number K, indicating the number of portals.

Then K lines follow. Each line has two numbers a_i and b_i, indicating that you place a portal at position a_i and it teleports the frog to position b_i.

You should guarantee that $1 \leq K, a_i, b_i \leq 199$, and $a_i \neq a_j$ if $i \neq j$. If there are multiple solutions, any one of them is acceptable.

Sample Input

0
1
5

Sample Output

2
1 1
2 1
2

1 199
2 2
2
4 199
5 5

Problem E. Xor

Description

You are given 2^k numbers: $a_0, a_1, \ldots, a_{(2^k-1)}$.

Define $s(x)$ as the following:

$$s(x) = \begin{cases} 0 \text{ if there are even bits of 1 in x's binary representation} \\ 1 \text{ otherwise} \end{cases}$$

You need to support five types of operations:

- output $Xor_{i=l}^{r} s(i)a_i = s(l)a_l \; xor \; s(l+1)a_{(l+1)} \; xor \; \ldots xor \; s(r)a_r$.
- Assign $a_{(i \; xor \; x)}$ to a_i simultaneously for each i in interval $[l, r]$.
- Assign a_i to $a_{(i \; xor \; x)}$ simultaneously for each i in interval $[l, r]$.
- Assign $(a_i \; xor \; x)$ to a_i simultaneously for each i in interval $[l, r]$.
- Assign y to a_x.

Xor here refers to bitwise operation *Exclusive Or*. In binary representation, the result in each position is 1 if the bits in two factors are different, and 0 if they are the same. For example,
$5 \; xor \; 3 = 6$, which is $(101)_2 \; xor \; (011)_2 = (110)_2$

Input

The first line contains an integer T, indicating the number of test cases.
For each test case:
The first line contains an integer k $(0 \le k \le 17)$.

The following line contains 2^k integers, the i-th integer is $a_{(i-1)}$ $(0 \le a_i < 2^k)$.

The next line contains an integer Q $(1 \le Q < 2^k)$, indicating the number of operations.

Then next Q lines, each line is one of:

- $1 \; l \; r$: output $Xor_{i=l}^{r} s(i)a_i$.

- 2 x l r: Assign $a_{(i\ xor\ x)}$ to a_i simultaneously for each i in interval $[l, r]$.
- 3 x l r: Assign a_i to $a_{(i\ xor\ x)}$ simultaneously for each i in interval $[l, r]$.
- 4 x l r: Assign $(a_i\ xor\ x)$ to a_i simultaneously for each i in interval $[l, r]$.
- 5 x y: Assign y to a_x.

It is guaranteed that $0 \le x, y < 2^k$ and $0 \le l \le r < 2^k$.

Output

For each operation 1, output a line containing a integer, indicating the answer.

Sample Input

```
1
3
7 6 1 3 1 7 2 4
5
1 3 3
2 4 3 7
5 2 6
1 2 5
1 1 6
```

Sample Output

```
0
1
7
```

Problem F. The Kth Largest Value

Description

Given a directed graph with n vertices and m edges, pair (u,v) is good if and only if there is a path from vertex u to vertex v. Obviously, pair (u,u) is also good.

The value of a good pair (u,v) equals to u xor v. Xor here refers to bitwise operation *Exclusive Or*.

We will query Q times. Each time we give an integer K, please print the Kth largest value among all the good pairs.

Input

The first line of input is an integer T(1 <= T <= 3) , the number of test cases.

For each test case:

The first line contains three integers n(1<= n <= 50000) ,m(1 <= m <= 200000) and Q(1 <= Q <= 10), meaning that there are n vertices and m edges in the graph, and there are Q queries

Then m lines follow. Each line includes two integers u and v (1 <=u , v <= n), standing for a directed edge from u to v. The graph won't contain multiple edges between the same pair of vertices.

Then Q lines follow. Each line includes an integer, above mentioned K (1 <= K <= 10^9).

It is guaranteed that the answer always exists.

Output

For each test case, print the Kth largest value of all good pairs.

Sample Input

```
2
3 3 3
1 2
1 3
2 3
1
2
3
7 7 5
1 2
3 5
5 4
4 3
1 5
2 6
5 7
```

1
3
5
7
9

Sample Output

3
2
1
7
7
6
5
4

Problem G. Solving Equations is Easy

Description

Alice and Bob are good friends. After a math class about solving equations, they have a little conversation.

Alice : Solving equations is easy.

Bob : I don't believe it. What's the roots of $x^2 - x - 2 = 0$?

Alice : Too simple. -1 and 2 are the roots.

Bob : How about $x^3 + 2x - x + 1 = 0$?

Alice : The real root is $\frac{\sqrt[3]{2(\sqrt{93}-9)}-2\sqrt[3]{\frac{3}{\sqrt{93}-9}}}{6^{\frac{2}{3}}}$. The complex roots are also easy to solve.

Bob : Unbelievable! You are really good at solving equations. Now I have a challenge for you. Consider the equations $x^2 - x - 2 = 0$, can you construct a new quadratic equations, whose roots are $(-1)^2$ and 2^2 ?

Alice : (After few seconds) It is $x^2 - 5x + 4 = 0$.

Bob : Excellent.

When Bob gets back to dormitory, he asks his roommates, YOU, the same questions. Since you have a computer and you are good at programming, you need to solve a more challenging problem.

Given an integer m and an equations $x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$, where the roots (including real roots and complex roots) are $x_1 \ldots x_n$, can you construct a new equations $y^n + b_{n-1}y^{n-1} + \cdots + b_1y + b_0 = 0$, whose roots are $y_1 \ldots y_n$ and satisfy $y_1 = x_1^m, \ldots, y_n = x_n^m$?

Input

There are at most 15 test cases.

In each test case, there are two lines of input.

The first line contains two integers, n and m.

The second line contains n integers, $a_0 \ldots a_{n-1}$, which are the coefficients of the equations $x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$.

The input ends with one line of two 0s.

The data ensures that $n, m \geq 1$, $n \leq 6$, $n + m \leq 10, |a_i| \leq 120$.

Output

For every test case, output one line of integers, $b_0 \ldots b_{n-1}$, which describes the equation $y^n + b_{n-1}y^{n-1} + \cdots + b_1 y + b_0 = 0$.

The data ensures that $|b_i| < 10^{12}$.

Sample Input

1 9
2
2 2
2 -3
2 2
2 -1
0 0

Sample Output

512
4 -5
4 3

Explanation

In the first test case, Bob gives you $x + 2 = 0$, where the root is -2. The corresponding equation is $x + 512 = 0$, whose root is -2^9.

In the second test case, Bob gives you $x^2 - 3x + 2 = 0$, where the root are 1 and 2. The corresponding equation is $x^2 - 5x + 4 = 0$, whose roots are $1^2 and\ 2^2$.

In the third test case, Bob gives you $x^2 - x + 2 = 0$, where the roots are $\frac{1}{2}(1 \pm \sqrt{7}i)$.

The corresponding equation is $x^2 + 3x + 4 = 0$, whose roots are $\frac{1}{2}(-3 \pm \sqrt{7}i)$.

Problem H. Approximate Matching

Description

String matching, a common problem in DNA sequence analysis and text editing, is to find the occurrences of one certain string (called pattern) in a larger string (called text). In some cases, the pattern is not required to be exactly in the text, and minor differences are acceptable (due to possible typing mistakes). When given a pattern string and a text string, we say pattern P is approximately matched within text S, if there is a substring of S which is at most one letter different from P. Note that the length of this substring and the pattern must be identical. For example, pattern “abb” is approximately matched in text “babc” but not matched in “bbac”.

It is easy to check if a pattern is approximately matched in a text. So your task is to count the number of all text strings of length m in which the given pattern can be approximately matched, and both of the patterns and texts are binary strings in order not to handle big integers.

Input

The first line of input is a single integer T ($1 \le T \le 666$), the number of test cases. Each test case begins with a line of two integers n,m ($1 \le n,m \le 40$), denoting the length of pattern string and text string. Then a single line of binary string P follows, which denotes the pattern. Note that there will be at most 15 test cases in which $n \ge 16$.

Output

For each test case, output a single line with one integer, representing the answer.

Sample Input

```
5
3 4
110
4 7
1011
2 10
00
7 17
1001110
11 22
11101010001
```

Sample Output

```
12
104
1023
72840
291544
```

Problem I. Palindromes

Description

Recently, Nvoenewr learnt palindromes in his class.

A palindrome is a nonnegative integer that is the same when read from left to right and when read from right to left. For example, 0, 1, 2, 11, 99, 232, 666, 998244353353442899 are palindromes, while 10, 23, 233, 1314 are not palindromes.

Now, given a number, Nvoenewr can determine whether it's a palindrome or not by using loops which his teacher has told him on the class. But he is now interested in another question: What's the K-th palindrome? It seems that this question is too difficult for him, so now he asks you for help.

Nvoenewr counts the number from small to big, like this: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 22, 33, 44, 55, 66, 77, 88, 99, 101 and so on. So the first palindrome is 0 and the eleventh palindrome is 11 itself.

Nvoenewr may ask you several questions, and the K may be very big.

Input

The first line contains one integer T(T <= 20)——the number of questions that Nvoenewr will ask you.

Each of the next T lines contains one integer K. You should find the K-th palindrome for Nvoenewr.

Let's say K is a n-digit number. It's guaranteed that K>=1, 1<=n<=100000 and the sum of n in all T questions is not greater than 1000000.

Output

Print T lines. The i-th line contains your answer of Nvoenewr's i-th question.

Sample input

4
1
10
11
20

Sample Output

0
9
11
101

Problem J. Rikka with Triangles

Description

Rikka places n points on a two-dimensional Cartesian coordinate system. The coordinate of the ith point is (x_i,y_i) . She wants to choose three points to make a weapon, and then she will use it to fight against the administration of "invisible boundaries".

To make this weapon more powerful, Rikka restricts the shape of the triangle to acute triangles. There may be several legal plans to make the weapon. Rikka wants you to calculate the sum of the areas of the triangles among all these plans.

Two plans are considered different if and only if they differ in at least one point.

NOTE: You can use __int128 in your program.

Input

The first line contains a single integer T(1 <= T <= 5), the number of the test cases.

For each test case, the first line contain a single integer n(3 <= n <= 2000) , the number of the points. And then n lines follow, each line contains two integers x and y ($|x|,|y| <= 10^{18}$), the coordinates of a point. The input guarantees that no two points are identical.

Output

Let S be the sum of the areas, it can be proved that 2×S is an integer.

Since the answer may be very large, you only need to output 2×S modulo 998244353 (a prime number).

Sample Input

```
3
3
1 1
2 2
2 3
3
1 1
2 3
3 2
4
1 1
3 1
4 1
2 3
```

Sample Output

```
0
3
10
```